

SW2

EventLens: Event Risk Research for Indices and Equities

by

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Abstract

EventLens is a proposed research dashboard that connects prediction-market events to the risks faced by investors in major US indices and a small selection of popular equities. A user will be able to find relevant events, compare the market's probability with a wallet-informed estimate, explore possible portfolio outcomes, and ask an AI assistant to explain the supporting evidence.

The central research question is whether a wallet's earlier forecasting record adds useful information beyond Polymarket prices and overall trading activity. We will combine market data, public blockchain records and equity-price histories, using only information available at each historical forecast time. Simple probability models will be evaluated first, followed by event-based portfolio scenarios and volatility forecasts. The initial event family will be US Federal Reserve rate decisions, subject to a data-feasibility review.

The prototype will cover three index-tracking exchange-traded funds and five selected equities, with cash available for paper portfolio comparisons. AI agents will retrieve evidence, request calculations from tested services and explain results. Numerical estimates will remain traceable to their data and model versions. Historical testing, data quality and user comprehension will determine success; improved prediction or profitable trading will be tested rather than assumed.

Table of Contents

1	Introduction	1
1.1	Background and Motivation	1
1.2	Objectives	1
1.3	Literature Survey	3
2	Methodology	4
2.1	Design	4
2.2	Implementation	12
2.3	Testing	13
2.4	Evaluation	13
3	Project Planning	15
3.1	Division of Work	15
3.2	GANTT Chart	15
4	Hardware and Software Requirements	16
4.1	Hardware Requirements	16
4.2	Software Requirements	16
5	References	17
6	Appendix A: Meeting Minutes	19

1 Introduction

1.1 Background and Motivation

Financial assets are repeatedly repriced around discrete events such as central-bank decisions, elections, and geopolitical developments. Investors can observe news, asset prices, and volatility indicators, but these sources do not directly show a continuously updated probability of the underlying event. Prediction markets such as Polymarket aggregate the beliefs of market participants into continuously updated probabilities, which can serve as a useful predictor of the real-world events [4].

Beyond these aggregate probabilities, Polymarket adds a distinctive Web3 dimension through a hybrid trading architecture. Orders are created and matched through an off-chain central limit order book, while matched trades settle on Polygon. Its blockchain-based infrastructure provides transparent transaction histories that enable the behaviour and historical performance of individual participants to be analysed. This creates an opportunity to investigate whether information embedded in prediction-market activity can provide useful signals for traditional financial markets.

This project aims to develop a machine-learning-assisted quantitative analysis framework that extracts information from Polymarket and evaluates its value in forecasting financial asset volatility.

Research question

Could information embedded in Polymarket's on-chain trading activity improve conventional financial-market volatility forecasts and provide actionable insights for investors?

1.2 Objectives

1. **Develop a smart-wallet identification model** using historical Polymarket on-chain trading behavior to distinguish informative traders from noise, hedging, and other non-predictive activity.
2. **Construct adjusted event probabilities** by weighting prediction-market information according to the estimated informativeness of smart wallets, and compare it with raw probability inferred from the bets.
3. **Develop a volatility forecasting model** that incorporates Polymarket-derived signals alongside a conventional market volatility baseline.
4. **Evaluate whether Polymarket data provides incremental predictive value** through historical backtesting and comparison against benchmark volatility forecasts, and present the results through an interactive AI-assisted dashboard.

The primary target user is a research-oriented retail investor or finance professional who understands basic portfolio concepts but lacks the infrastructure to query blockchains, evaluate prediction-market participants, and construct event-conditioned risk models. EventLens is intended as a research and paper-decision tool rather than personalised financial advice.

Scope

This project will be separated in two phases. The first phase will focus on the minimum viable product (MVP), to validate the core concept, while the second phase will explore extensions and additional features. The scope of the project is summarized in Table 1, which outlines the minimum viable scope and the extended scopes.

Table 1: Minimum viable scope and extended scopes.

Dimension	MVP commitment	Extensions
Venue	Polymarket on Polygon	Cross-platform comparison, for example from Kalshi
Event coverage	Fed rates decisions	Other Macro and geopolitical events
Assets	Indices and ETFs	User-defined multi-asset portfolio
Horizons	1, 5 and 20 trading days	Additional term-structure horizons
Wallet model	Address-level, past-only scoring	Validated entity clustering

1.3 Literature Survey

Prediction markets and informative participants

Research on prediction markets suggests that prices may be disproportionately influenced by a relatively small group of informed or disciplined participants. Oliven and Rietz [9], for example, find that active market makers in the Iowa Electronic Markets behave more consistently with rational pricing conditions than typical price takers. This supports the “marginal trader” hypothesis: market accuracy may depend less on the average participant than on the traders willing and able to act against mispricing. Evidence from forecast aggregation is more mixed. Chen et al. [3] find that weighting forecasters according to past performance does not significantly outperform simpler pooling methods in their sports-prediction sample. By contrast, Atanasov et al. [1] show that performance weighting, temporal decay and recalibration can produce forecasts that outform standard prediction-market prices.

Recent Polymarket research provides more direct evidence of participant heterogeneity. Akey et al. [2] find that approximately 69% of users lose money and that the top 0.1% capture more than half of aggregate gains. Gómez-Cram et al. [8] classify approximately 3% of accounts as skilled. These accounts do not hold insider information but exhibit out-of-sample persistence and respond more rapidly to public information.

Prediction-market price as a proxy of event probability

Binary prediction-market contracts pay one unit if a specified outcome occurs and zero otherwise. Their prices are therefore commonly interpreted as event probabilities. Wolfers and Zitzewitz [4] argue that this interpretation can be a useful approximation under risk neutrality, limited risk aversion and sufficiently liquid trading. Nevertheless, the observed price remains an equilibrium outcome determined by traders’ beliefs, wealth, preferences and the market mechanism, not automatically the physical probability that the event will occur.

Manski [14] demonstrates this limitation formally. With heterogeneous risk-neutral price-taking traders, the equilibrium price represents a budget-weighted quantile of the belief distribution rather than its mean. Consequently, the price may bound the average belief without identifying it exactly. Gao et al. [16] extend the analysis to heterogeneous risk-averse traders and utility-based market makers. Depending on the utility specification, the limiting price may resemble a geometric or power mean of beliefs and also depend on risk aversion, initial wealth and trading behaviour.

2 Methodology

2.1 Design

A guided research journey

A user selects a supported asset or paper portfolio and a time horizon. EventLens presents relevant events, compares market and wallet-informed probabilities, and lets the user investigate the underlying wallet activity before exploring portfolio scenarios. A contextual AI assistant explains the evidence, and saved reports preserve the analysis for later review.

For example, a user researching SPY over five trading days could inspect an upcoming rate decision, identify which tracked wallets are increasing YES or NO exposure, and compare the resulting risk scenarios. An incompatible announcement horizon or unsupported asset-response estimate will be explained before withholding the affected calculation.

Table 2: Core interface features and the decisions they support.

Screen or feature	What the user sees	What the user can do
Overview and watchlist	Supported assets, upcoming eligible events, source freshness and coverage.	Select holdings, choose a horizon and open an event.
Event detail	Exact question, resolution rule, announcement time, raw and adjusted probability history.	Compare the estimates and inspect uncertainty and liquidity.
Smart-wallet tracker	Tracked addresses, prior event history, YES/NO exposure, recent activity and evidence quality.	Follow wallets, inspect their records and compare their contribution to an event signal.
Scenario studio	YES/NO outcomes, portfolio loss estimates, VaR, expected shortfall and response assumptions.	Change a reviewed stress assumption or compare a bounded paper allocation.
Research assistant	An explanation tied to the selected event, portfolio, horizon and evidence.	Ask follow-up questions and open the supporting sources.
Saved research	Analysis date, inputs, model and scenario versions, source links and limitations.	Revisit a frozen run and export a readable report with a machine-readable result.

Smart-wallet tracking and evidence inspection

Users can browse wallets in covered events, filter by history and activity, and follow public addresses in a research watchlist without connecting a wallet. Profiles show earlier resolved-event decisions, entry prices, subsequent price movements, reconstructed positions and an activity timeline. A provisional “smart” label denotes historically informative behaviour, supported by a dated score, sample size and coverage assessment.

Within an event, users can compare wallets’ YES/NO exposure changes, agreement and concentration, then inspect their contributions to the adjusted estimate. Personal watchlists do not change model-selection rules. Source and blockchain transaction links support inspection, with sampled verification distinguished from unchecked records.

Results show their horizon, as-of time and missing evidence, separating observations from estimates and assumptions. Users can request explanations and save analyses. Insufficient wallet history leaves the adjustment unavailable while retaining supported market information.

Concept screens

Figures 1 and 2 illustrate the intended layout. They are conceptual interface images; values are illustrative and do not represent research results.

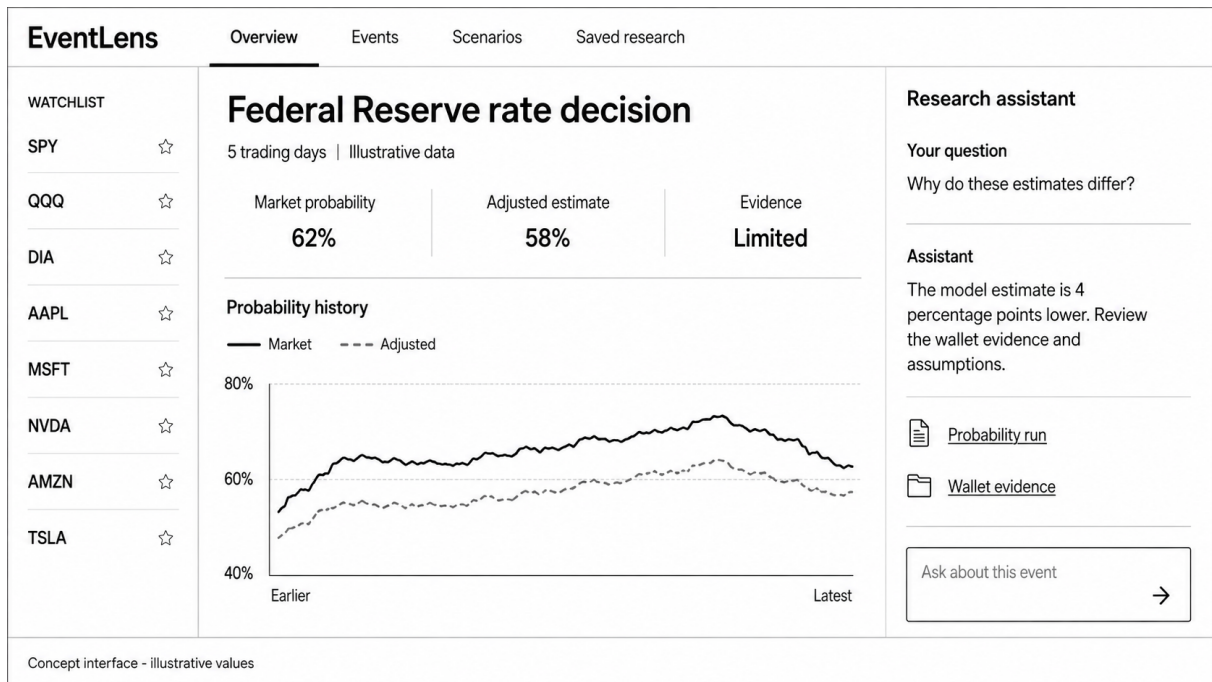


Figure 1: Overview concept: a watchlist, event probability comparison and contextual research assistant. All displayed values are illustrative.

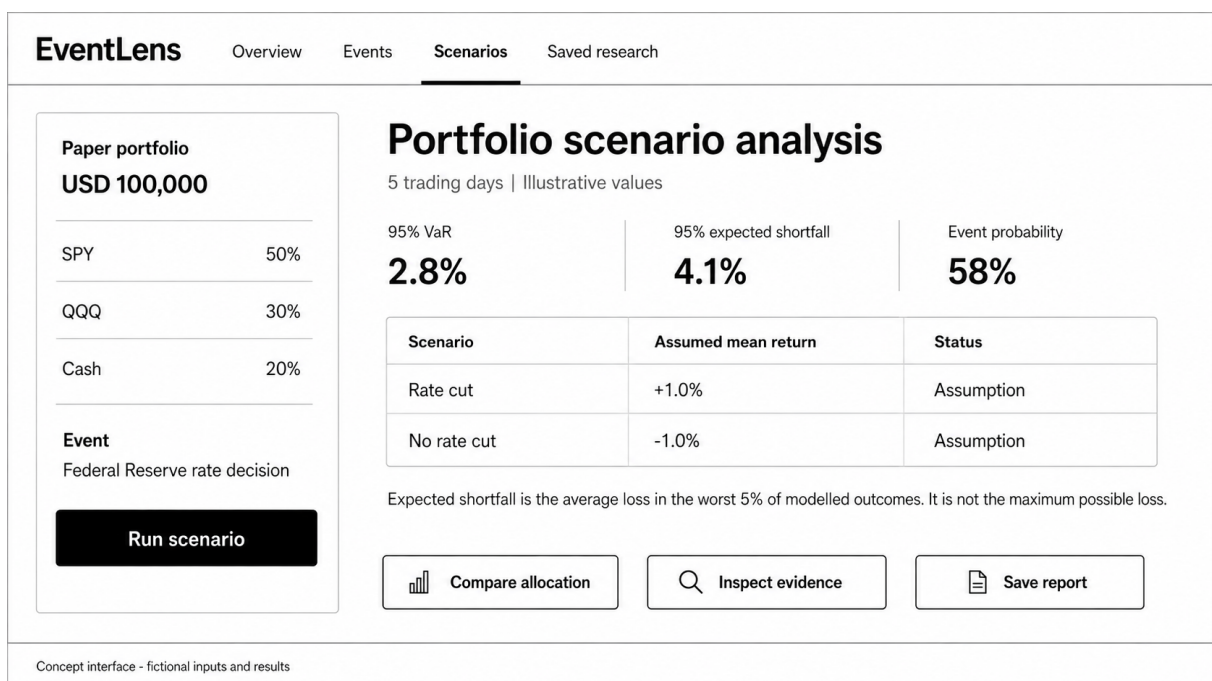


Figure 2: Scenario concept: portfolio inputs, clearly labelled model outputs and visible assumptions. The user compares paper scenarios without placing orders.

Three roles within one controlled workflow

The proposed assistant will use three logical roles: an **event research agent** to find relevant evidence, an **analysis coordinator** to request approved calculations, and an **explanation agent** to present results in context. These roles can share one model and orchestration service in the MVP. A separate autonomous agent infrastructure is unnecessary.

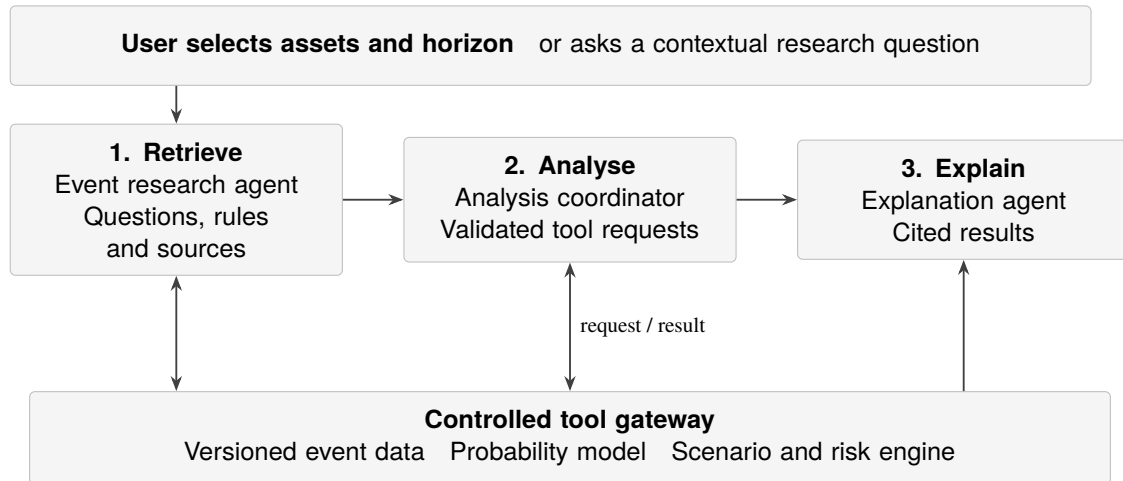


Figure 3: Proposed agent workflow. All quantitative results come from the same validated services used by the dashboard.

From a question to a supported answer

For “How could the next rate decision affect my SPY and MSFT exposure?”, the assistant resolves the assets, horizon and analysis time. It retrieves an eligible event and proposes an event-to-asset explanation. Numerical responses must come from historical estimates or reviewed stress assumptions; generated text alone is not sufficient evidence.

The coordinator can call tools such as `find_events`, `get_wallet_evidence`, `get_probability`, `run_scenario` and `compare_allocations`. The gateway checks asset coverage, weights, units, data freshness and horizon compatibility. It returns structured values with sources, model versions and a research-run identifier. The explanation then links its numerical claims to those results and separates observations, estimates and assumptions [12, 13].

Boundaries and review

Users confirm changes to their paper inputs and can inspect assumptions before running a comparison. Researchers review new event-to-asset mappings and choose model versions using validation results. The agent cannot alter wallet labels, tune against the final test period, invent missing prices, calculate risk in prose or submit live orders. Market descriptions are treated as untrusted data, including any embedded instructions.

When evidence is incomplete, the assistant will identify the gap and offer a supported view, such as the raw probability. Tool traces and saved reports make answers reviewable. Evaluation will cover citations, numerical agreement, abstention, user understanding, response time and cost.

Sources and their roles

The data pipeline will combine three kinds of evidence: prediction-market information, public blockchain settlement records and conventional financial data. These sources serve different purposes. Polymarket matches orders off-chain and settles trades on-chain, so an order-book snapshot is not itself a blockchain record [15].

Table 3: Planned sources, with access decisions made during the feasibility audit.

Source	Data to acquire	Use and limitation
Polymarket metadata API (Gamma)	Event definitions, rules, identifiers, outcome prices and available price history.	Define the forecast target; archive changes to mutable metadata [16].
Polymarket CLOB and activity Data API	Bid/ask snapshots, liquidity and public wallet activity.	Measure market quality and candidate wallet flow; historical book depth may need prospective collection.
Goldsky Polymarket datasets	Indexed fills, matched trades and relevant position records.	Targeted historical reconstruction, subject to version coverage and access cost [17].
Polygon RPC	Transaction receipts, contract logs and block references.	Verify a stratified sample of settlements; older state may require archival access.
Alpha Vantage or an approved equivalent	Daily equity/ETF prices, splits and dividends; intraday data only if available.	Proposed primary price-provider adapter. Adjusted daily data are a premium function; coverage and academic-use terms must be checked [18].
Federal Reserve	Meeting dates, statements and announced rate decisions.	Establish announcement timing and validate event labels separately from platform resolution [19].

How on-chain data benefits EventLens

Public settlement records link activity to addresses and outcome tokens [15]. EventLens will use these histories to measure historically informative wallets' directional exposure, agreement and concentration across covered events. Separating transfers and token lifecycle operations from purchases avoids treating every balance increase as a new prediction.

Users can trace contributions to recorded transactions, and researchers can check indexed data against chain receipts. Addresses represent pseudonymous accounts, not verified people or independent decision-makers. Comparing models with and without wallet features will test whether this Web3 evidence improves forecasts; provenance alone does not establish skill.

A bounded collection strategy

We will backfill only eligible events and relevant wallet histories, using indexed data and hosted RPC without running a blockchain node. Goldsky warns that legacy subgraphs may be incomplete after the 2026 contract migration; the audit will verify version-specific coverage [17].

The audit will test sample downloads and fix provider, adjustment convention and coverage for each experiment. Forward market snapshots will be collected from project start and equity bars refreshed after each session. Permitted university data are a fallback. Access and historical depth remain unconfirmed; options quotes are optional.

Turning source records into research inputs

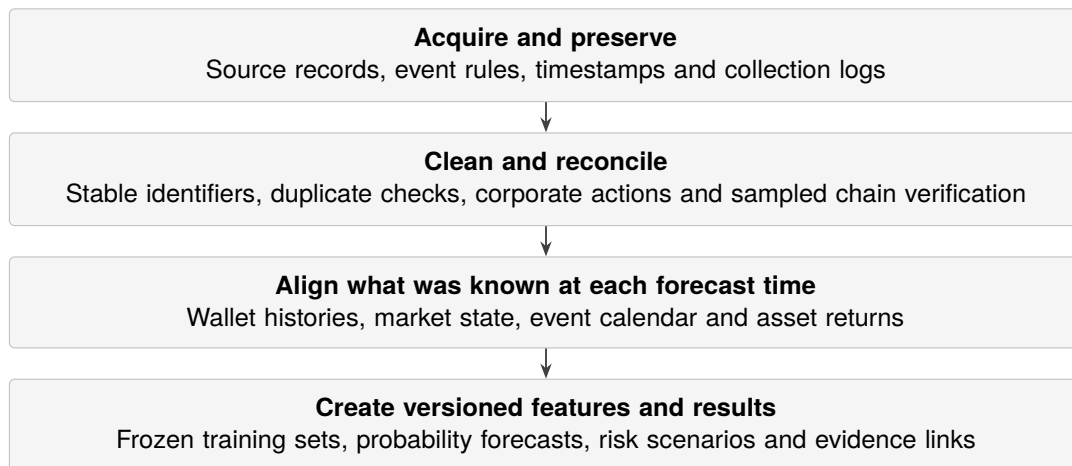


Figure 4: Processing pipeline. Raw evidence is retained so each transformation can be checked and repeated.

Preserve identifiers and time. Each event is linked to its markets, outcome tokens and settlement records. Raw records retain source time, collection time and source version. Announcement time, platform resolution time and data-availability time remain separate. Historical records downloaded later are admissible only when their content can be shown to have existed at the forecast cut-off; mutable current rankings are not historical features.

Reconstruct activity consistently. Wallet analysis retains both participant perspectives, while market volume counts the economic trade once. A transaction may contain several logs, so transaction hash alone is not a unique record key. We will use chain, transaction, log and contract identifiers and verify sampled amounts against Polygon receipts. Position reconstruction must account for transfers, splits, merges and redemptions as well as trades. Incomplete positions receive a quality flag.

Align asset returns. Timestamps will be normalised to UTC while retaining the US trading calendar, daylight-saving changes and after-hours announcements. Splits and dividends will be handled consistently. Future adjustments or revised data must not change historical features without a recorded version. Missing bars will be flagged rather than filled with zero returns; stale probabilities will not be carried across an announcement as if newly observed.

Make processing reproducible. Immutable source snapshots and checksums will accompany cleaned tables and Parquet research files. Collectors will use retries, rate limits and checkpoints. Replaying the same batch must not duplicate records. Each model run stores the data manifest, configuration, code version and random seed, and the interface will expose coverage gaps through its evidence panel.

Estimate the event probability

We will begin with a raw probability based on the YES bid/ask midpoint, recording spread, depth and timestamp. When both outcome books are usable, their midpoints can be normalised for consistency. A missing or one-sided book will be flagged, not silently replaced by an unrelated latest trade.

A simple regularised logistic model will calibrate this estimate using earlier resolved events. The wallet-informed model will then add a small number of signals: overall directional flow, the flow of historically informative wallets and market-quality controls. Coefficients will be fitted on training data, rather than chosen to produce a desired adjustment. Beta calibration is a possible robustness check if justified by the sample [14].

Select and track informative wallets

An informative wallet is an observed account whose earlier decisions contain information beyond the contemporaneous market price. Our first score will examine whether the wallet repeatedly chose the eventual outcome at informative prices, whether prices subsequently moved in its direction, and how broad its prior history was. Returns after fees will be a supporting diagnostic; profitable liquidity provision alone is not proof of forecasting skill.

Repeated fills will be consolidated into wallet–event decisions. Balanced YES and NO holdings will be distinguished from directional exposure. Scores based on few independent events will be pulled towards the population average, a technique called shrinkage. A provisional eligibility rule is ten earlier resolved events across at least three announcement dates, with five- and twenty-event thresholds checked in sensitivity analysis. If the history cannot support this rule, the wallet signal will be unavailable rather than assigned artificial confidence.

Wallet eligibility and scores will use only records available and outcomes resolved by each forecast cut-off. New activity updates exposure; newly resolved events update later scores. Frozen snapshots prevent today's successful wallets from being retrospectively selected for earlier tests.

Contributions combine earlier skill, direction, recency and capped size so one large wallet cannot dominate. Comparing total and eligible-wallet flow tests whether selection adds information. Profiles expose these components, timestamps, concentration and disagreement; new-wallet anomalies remain separate from historical skill.

Link events to assets

The event research agent may suggest that a rate decision matters to SPY or MSFT, but numerical impacts will come from matched historical event windows and conventional market controls. The mapping must account for surprise relative to the expectation already embedded in prices. Sparse asset-specific estimates will be pooled or replaced by explicitly reviewed stress ranges.

Wallet information may improve an event probability without improving an equity-risk forecast. The event-to-asset response is a separate estimate with its own uncertainty and evidence requirements.

Each mapping will retain its event definition, horizon, sample size, estimated response range and review status. A terminal YES/NO mixture will only be used when the economic announcement falls within the risk horizon. Otherwise the MVP will offer a compatible horizon or withhold that calculation. Correlated contracts from one meeting will not be treated as independent shocks.

Explore portfolio scenarios and tail risk

For a selected portfolio, the engine will combine the conditional return distributions for an event occurring (YES) and not occurring (NO). Let p be the selected event probability, μ a mean portfolio return and σ^2 its variance over the same horizon. The key calculations are:

$$\mu_{\text{mix}} = p\mu_{\text{YES}} + (1 - p)\mu_{\text{NO}}, \quad (1)$$

$$\sigma_{\text{mix}}^2 = p\sigma_{\text{YES}}^2 + (1 - p)\sigma_{\text{NO}}^2 + p(1 - p)(\mu_{\text{YES}} - \mu_{\text{NO}})^2. \quad (2)$$

The last term captures uncertainty about which outcome occurs. Consequently, a higher adverse-event probability can worsen expected loss without necessarily increasing variance. For multiple holdings, we will preserve cross-asset dependence when sampling returns, then apply the user's weights to each joint scenario.

The engine will report scenario returns, horizon volatility, a chosen loss-threshold probability, 95% VaR and 95% expected shortfall. VaR is a threshold, not the maximum loss. Expected shortfall will be calculated as the average of the worst 5% of the weighted loss distribution, including fractional threshold mass where needed. We will use empirical or heavy-tailed scenarios and check sensitivity to the probability and asset-response assumptions.

Paper comparisons will hold the event and model version constant while comparing the current allocation with a user-selected reduction in a supported holding and a corresponding increase in cash. Weights must sum to one and remain non-negative. Any displayed benefit must be considered alongside turnover, transaction-cost assumptions and estimation uncertainty.

Volatility Forecasting & Delta-Hedged Trading Strategy Extension

Once the on-chain-adjusted probability p_{adj} is obtained, EventLens bridges prediction-market dynamics to traditional asset volatility through a dedicated Volatility Forecasting and Delta-Hedged Trading Strategy extension. This pipeline automates contract selection, feature extraction, volatility model fitting, and delta-hedged paper trading to evaluate the forecasting performance of on-chain signals.

Recurring event series and contract selection Polymarket serves as the sole source of prediction-market data for EventLens. The platform provides continuous tracking across recurring event families, including macroeconomic releases (e.g., FOMC rate decisions, CPI releases) as well as binary Polymarket contracts on equity index closing levels (e.g., S&P 500 / NASDAQ daily and weekly target closes) and precious metals price targets (e.g., Gold and Silver closes).

To maintain temporal consistency across continuous volatility forecasting horizons, an automated contract selection engine identifies and locks onto the closest-to-expiry market contract for any chosen event family. When event resolutions overlap or liquidity is fragmented across multiple contracts, the selection engine filters for active contracts meeting minimum volume and bid-ask spread thresholds, continuously rolling to the next active contract upon resolution.

Feature engineering pipeline The feature extraction layer constructs point-in-time predictive signals derived from the adjusted probability p_{adj} alongside raw market microstructure variables. Users are given interactive control to toggle and configure exogenous inputs depending on their desired baseline.

- **Probability Signal Metrics:** Includes p_{adj} , raw probability p_{raw} , probability spread ($p_{\text{adj}} - p_{\text{raw}}$), model

confidence bands, and probability jump/surprise signals (Δp_{adj} measured over historical lookback windows up to the forecast cutoff).

- **User-Configurable Exogenous Baselines:** Allows optional inclusion of benchmark volatility proxies (e.g., VIX, MOVE) and historical realized volatility as regime controls.

Realized volatility forecasting models The extracted signals serve as exogenous features to forecast future realized volatility over standard horizons (e.g., 1-day, 5-day). The core target over H trading days is defined as the sum of squared daily log returns:

$$RV_{t,H} = \sum_{h=1}^H r_{t+h}^2. \quad (3)$$

EventLens implements three core volatility model structures:

- **HAR-RV-X (Heterogeneous Autoregressive with Exogenous Signals):** Extends the classical HAR-RV framework (which models realized volatility via daily, weekly, and monthly components) by adding linear exogenous terms for p_{adj} and probability jump signals. Assumes a linear, multi-frequency structure for volatility persistence.
- **GARCH-X:** Incorporates exogenous prediction-market jump and log-odds signals directly into the conditional variance equation of a standard GARCH model. Assumes stationarity and clustering in conditional volatility returns.
- **Machine Learning Baselines (XGBoost / Random Forest):** Non-linear tree ensemble models fitted on historical volatility features, probability spreads, and regime signals. Evaluated using standard robust loss functions such as QLIKE (Quasi-Likelihood) and Mean Squared Error (MSE) to prevent extreme volatility outliers from distorting model fitting.

Delta-hedged backtesting framework To demonstrate the practical utility of the forecasted realized volatility, EventLens runs an automated delta-hedged backtest. The backtest engine compares the model's forecasted realized volatility against current option-implied volatility (IV) from option market quotes to identify implied-versus-forecasted variance gaps.

When a variance gap is detected, the strategy simulates a delta-neutral position (e.g., ATM options hedged continuously against the underlying asset) to harvest the variance risk premium. Costs, bid/ask spreads, hedge frequency, and execution assumptions are included. This simplified backtest framework provides the user with an intuitive evaluation of forecast performance, displaying backtested Sharpe ratios, maximum drawdowns, and simulated trade logs directly on the research dashboard.

2.2 Implementation

We propose a modular Python backend and a TypeScript web frontend. One deployable application with a background worker will keep the four-person project manageable while separating collection, modelling, risk calculation and AI explanation.

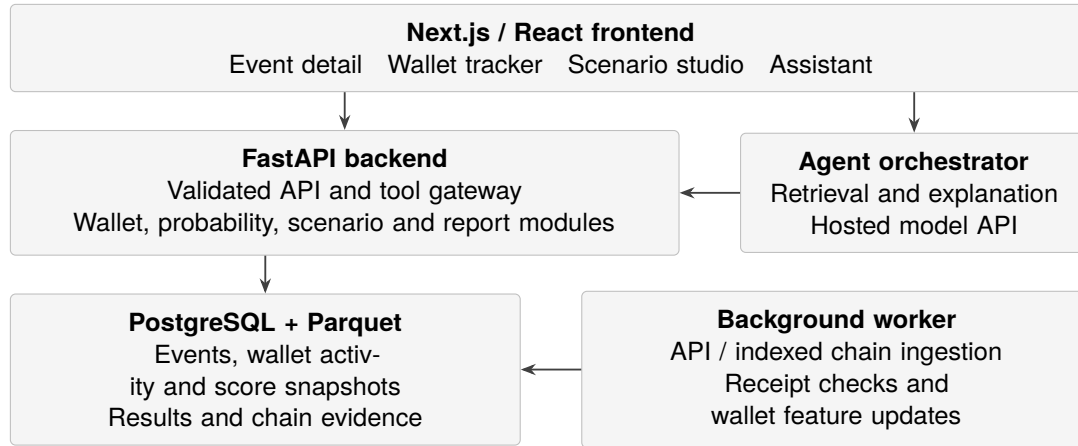


Figure 5: Proposed application architecture. The dashboard and assistant share the same calculation services and stored results.

Application services

Next.js and React will provide interactive views; FastAPI and Pydantic will validate requests and share calculation services with the assistant [21, 20]. Polars/Pandas, NumPy and scikit-learn will support research. PostgreSQL will hold linked records and run metadata, while Parquet preserves raw evidence and research datasets. A Python worker and Docker Compose will support one manageable deployment.

Smart-wallet tracking pipeline

The worker will discover candidate addresses from covered markets and incrementally collect their relevant histories through activity APIs and version-compatible indexed chain data. It will map condition and outcome-token identifiers, deduplicate logs, and reconcile fills with transfers, splits, merges and redemptions. Checkpoints and retries make collection resumable. Polygon receipt checks will verify sampled records; block references and verification status will accompany the evidence.

Separate tables will store wallet activity, position snapshots, dated skill scores, event-level contributions and user watchlists. Scores will be recomputed when eligible outcomes resolve; new activity will refresh exposure and aggregate flow. Each snapshot records its information cut-off, coverage and model version. Incomplete histories are flagged and excluded from calculations that require complete positions.

Backend endpoints will supply wallet search, profiles, watchlist updates, activity timelines and event contributions to both the tracker and assistant. The frontend will show the latest completed snapshot and refresh status. Watching an address does not make it model-eligible; scoring follows the shared research rules.

Long analyses return a job identifier and status. Cached results include data, wallet-score, model, scenario, portfolio and horizon versions. Authenticated team access, server-side secrets, persistent volumes and backups protect the deployment. Hosted-model prompts and versions will be recorded, and direct research remains available when the assistant is unavailable.

2.3 Testing

Testing checks that the implementation is *correct*, that the code and pipeline do what they are meant to do – independently of whether the underlying research hypothesis turns out to be true.

Data and pipeline tests. Ingestion will be checked for schema validity, duplicate-free replay after an interrupted collection run, and reconciliation of a sample of collected records against source APIs and blockchain data. A point-in-time check will reject any feature or wallet history that uses information published after its forecast cut-off, since a lookahead leak is the failure most likely to silently invalidate every downstream result without producing an obvious error. Corporate-action handling (splits, dividends) and timestamp normalisation across time zones and daylight-saving changes will also be covered, since these are easy to get subtly wrong and hard to notice later.

Calculation tests. The probability model, wallet-scoring routine and the scenario/risk engine will be unit-tested against reference cases with known outputs: event probabilities of exactly zero and one, a missing asset, a one-sided or illiquid order book, and a singular or near-singular covariance matrix. API-level tests will enforce structural invariants – portfolio weights non-negative and summing to one, horizons restricted to supported values, VaR and expected shortfall computed consistently with their definitions – so that a broken invariant fails loudly rather than producing a plausible-looking wrong number.

Workflow and agent tests. End-to-end tests will exercise the full path from event selection to a saved, shareable report, including stale or unavailable upstream data. Agent-specific tests will cover malformed or out-of-scope tool requests and adversarial instructions embedded in retrieved market or news text, since retrieved content is treated as untrusted data rather than instructions. The standing requirement is that every number shown to the user, including any number referenced by the AI assistant, traces back to a saved, versioned result rather than to generated text; this will itself be checked by a small set of tests that compare assistant output against the underlying stored result.

Indicative metrics. Pass/fail against the reference cases and invariants above; automated-test coverage of the ingestion and calculation modules; zero duplicate records after a replayed or interrupted collection run; and successful, byte-identical reproduction of a prior run’s results from its stored data manifest, configuration and random seed.

2.4 Evaluation

Evaluation asks whether the finished system meets the objectives in Section 1.2.

Table 4: How each objective will be evaluated (indicative; refined after the feasibility audit).

Objective	Question	Indicative metric
Wallet identification	Does the smart-wallet score capture information beyond chance?	Backtested accuracy/Brier score vs. a label-shuffled baseline; sensitivity to the eligibility threshold
Adjusted probability	Does wallet-weighting improve on the raw market price?	Brier score and log loss across the baseline ladder (raw price → +liquidity → +anonymous flow → +wallet signal)
Volatility forecasting	Does the augmented model beat a conventional baseline?	Variance error and QLIKE vs. historical/EWMA volatility; VaR exceedance rate
Dashboard and assistant	Is the system usable, and does it stay evidence-linked?	Formative user study: task completion, numerical agreement between assistant and dashboard, correct abstention on insufficient data

Experimental design

Comparisons will use a chronological train/validation/test split, with each event's contracts and snapshots kept together and overlapping return windows purged at split boundaries, so that no test-period information leaks backward. The final test interval stays unopened until models and thresholds are fixed. Wallet scores, calibration and scenario inputs will use only information available as of each historical forecast time, matching the point-in-time discipline enforced in testing.

Given the likely small number of independent announcement dates, results will be reported with confidence intervals rather than as single numbers, and a broader macro sample would require a documented scope revision rather than an informal expansion mid-project. Multiple wallet-level significance tests will use a false-discovery correction rather than being read individually. A null result – no measurable improvement from wallet or event signals – is a valid and reportable outcome; profitability is explicitly not a success criterion.

Usability evaluation

A small formative study (target 6–8 participants) will ask users to find an event, explain a probability difference, compare portfolio scenarios and identify a stated limitation, while we record task completion, time and errors. Fixed questions will compare three configurations – the full assistant, the dashboard alone, and a retrieval-only assistant without the calculation gateway – measured on source coverage, numerical agreement with the underlying stored result, correct abstention when evidence is insufficient, latency and estimated cost per query.

Risks to the evaluation itself

Table 5: Risks specific to producing a trustworthy evaluation, and planned responses.

Risk	Response
Too few independent events for reliable fitting	Report a descriptive pilot instead of a fitted comparison; widen the event family only via a documented revision.
Selection bias in "smart" wallets	Freeze the wallet universe and score cut-offs before the test window opens; apply shrinkage to low-sample scores.
Confounded asset response to an event	Use reviewed stress ranges and sensitivity analysis; keep event-probability and event-to-asset uncertainty separate.
Multiple-comparison inflation	Apply false-discovery control across wallet-level and baseline-ladder tests before reporting significance.
Unsupported AI claims in the user study	Restrict tool arguments, link every numeric claim to a stored result, and test missing-data abstention explicitly.

Success requires reproducible data and experiment configuration, a controlled baseline comparison, a tested risk engine, and a usable, evidence-linked interface; positive, negative and null findings will all be reported. Exact sample sizes, thresholds and the final participant count will be confirmed once the data-feasibility audit (Section 2.1) fixes what history is actually available.

3 Project Planning

3.1 Division of Work

Franklin, Anthony, Owen and Sunny will work across four tracks, each with a lead and a reviewer. Named ownership remains to be agreed. All members will contribute to literature review, reporting and presentations.

Table 6: Proposed work allocation, with named ownership to be recorded by the team.

Track	Responsibility	Deliverable
Data and blockchain	Source adapters, collection, reconciliation and quality reporting.	Versioned dataset and coverage report.
Probability research	Wallet scoring, calibration, baseline comparisons and leakage controls.	Reproducible probability experiments.
Scenarios and risk	Asset-response estimates, volatility, tail risk and reference tests.	Deterministic risk engine and application study.
Product and AI	Backend integration, frontend, agent tools and deployment.	Integrated dashboard and user evaluation.

Git will hold code, experiment configuration and separate LaTeX section files for collaborative editing. Generated files and secrets stay out of version control. Weekly reviews and an issue tracker will record owners, dependencies, meeting decisions and experiment changes.

3.2 GANTT Chart

The schedule is relative to project commencement and will be aligned with course dates. Data feasibility comes first; interface work starts with sample data to identify integration problems early.

Table 7: Planning baseline. Shaded cells indicate the main work period.

Workstream	1–4	5–8	9–12	13–16	17–20	21–24
Scope, literature and feasibility						
Collection and reconciliation						
Wallet and probability models						
Event mapping and risk engine						
Interface and AI integration						
Evaluation and user study						
Report and reproducibility						

Milestones. Week 2: initial access and scope decision. Week 7: a versioned dataset and reconciliation report. Week 12: first raw-price, anonymous-flow and wallet-flow comparison. Week 18: working scenarios and risk calculations. Week 22: integrated evaluation and user tasks. Week 24: final report, demonstration and reproducibility package. Failed feasibility gates will narrow the model or coverage before optional work begins.

4 Hardware and Software Requirements

4.1 Hardware Requirements

The initial models and API-hosted language model do not require a local GPU. Development can be carried out on standard personal computers, with a modest shared server for scheduled collection and the demonstration.

Table 8: Proposed hardware provision.

Resource	Minimum provision	Preferred provision
Developer computer	Four-core CPU and 8 GB RAM	16 GB RAM for concurrent local services.
Storage	30 GB free disk space	Approximately 100 GB SSD space for bounded historical data.
Shared server	Local demonstration if necessary	2–4 vCPU and 8–16 GB RAM, subject to measured workload.
Network	Stable internet connection	Reliable access for collectors, hosted models and remote collaboration.

A self-hosted Polygon node is outside the initial scope. Hosted RPC and targeted indexed data will be used for collection and verification. Database and raw-data storage must persist across application updates and be backed up regularly.

4.2 Software Requirements

Table 9: Required software and external services.

Category	Software or service	Purpose
Runtime	Python and Node.js	Research services, collectors and web application.
Application	FastAPI, Pydantic, Next.js, React and TypeScript	Validated backend APIs and interactive interface.
Data and modelling	PostgreSQL, Parquet, Polars/Pandas, NumPy and scikit-learn	Storage, processing and initial statistical models.
External services	Polymarket APIs, Goldsky, Polygon RPC, equity data and hosted LLM	Event evidence, asset histories and research assistance.
Quality and deployment	Pytest, browser tests, Docker and Docker Compose	Automated verification and reproducible deployment.
Collaboration	Git, issue tracker and LaTeX	Source control, task allocation and shared documentation.

External costs may include adjusted equity histories, indexed blockchain access, archival RPC, a hosted language model and server hosting. The team will set a spending cap after checking sample data and provider terms during the feasibility stage. A paid historical options dataset is not required for the core project. Software dependencies, model versions and configuration will be recorded to support reproducibility.

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6 Appendix A: Meeting Minutes

Initial project meeting

Date	8 August 2026
Location	Hong Kong
Attendees	Franklin, Anthony, Owen and Sunny
Purpose	Discuss the project direction, scope, evaluation approach and division of work.
Status	Preliminary discussion; formal scope decisions and individual ownership remained open.

Discussion summary

The meeting considered a Web3 quantitative-trading and market-analysis agent. Subsequent research narrowed the proposed direction to Polymarket, public wallet activity, event-conditioned volatility and portfolio risk. The present proposal focuses the initial application on selected indices and popular US equities, with options paper trading retained as a possible extension.

Matters requiring confirmation

The team will confirm the initial asset and event universe, named workstream leads, data access and the schedule against official course milestones. Decisions and action items will be recorded with owners and deadlines at the next meeting.

Formal scope approval and named ownership were not recorded at this meeting.